

Foundations of Medical image computing

[Medical image computing]

1.0 Content Block

Atlases Medical images can vary significantly across individuals due to people having organs of different shapes and sizes. Therefore, representing medical images to account for this variability is crucial. A popular approach to represent medical images is through the use of one or more atlases. Here, an atlas refers to a specific model for a population of images with parameters that are learned from a training dataset. The simplest example of an atlas is a mean intensity image, commonly referred to as a template. However, an atlas can also include richer information, such as local image statistics and the probability that a particular spatial location has a certain label. New medical images, which are not used during training, can be mapped to an atlas, which has been tailored to the specific application, such as segmentation and group analysis. Mapping an image to an atlas usually involves registering the image and the atlas. This deformation can be used to address variability in medical images.

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Studying temporal changes. Longitudinal studies acquire images over several months or years to study long-term processes, such as disease progression. Time series correspond to images acquired within the same session (seconds or minutes). They can be used to study cognitive processes, heart deformations and respiration. Combining complementary information from different imaging modalities. An example is the fusion of anatomical and functional information. Since the size and shape of structures vary across modalities, it is more challenging to evaluate the alignment quality. This has led to the use of similarity measures such as mutual information. Characterizing a population of subjects. In contrast to intra-subject registration, a one-to-one mapping may not exist between subjects, depending on the structural variability of the organ of interest. Inter-subject registration is required for atlas construction in computational anatomy. Here, the objective is to statistically model the anatomy of organs across subjects. Computer-assisted surgery. In computer-assisted surgery pre-operative images such as CT or MRI are registered to intra-operative images or tracking systems to facilitate image guidance or navigation. There are several important considerations when performing image registration:

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Diffusion MRI is a structural magnetic resonance imaging modality that allows measurement of the diffusion process of molecules. Diffusion is measured by applying a gradient pulse to a magnetic field along a particular direction. In a typical acquisition, a set of uniformly distributed gradient directions is used to create a set of diffusion weighted volumes. In addition, an unweighted volume is acquired under the same magnetic field without application of a gradient pulse. As each acquisition is associated with multiple volumes, diffusion MRI has created a variety of unique challenges in medical image computing. In medicine, there are two major computational goals in diffusion MRI:

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Mathematical methods in medical imaging A number of sophisticated mathematical methods have entered medical imaging, and have already been implemented in various software packages. These include approaches based on partial differential equations (PDEs) and curvature driven flows for enhancement, segmentation, and registration. Since they employ PDEs, the methods are amenable to parallelization and implementation on GPGPUs. A number of these techniques have been inspired from ideas in optimal control. Accordingly, very recently ideas from control have recently made their way into interactive methods, especially segmentation. Moreover, because of noise and the need for statistical estimation

techniques for more dynamically changing imagery, the Kalman filter and particle filter have come into use. A survey of these methods with an extensive list of references may be found in.

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Functional MRI Functional magnetic resonance imaging (fMRI) is a medical imaging modality that indirectly measures neural activity by observing the local hemodynamics, or blood oxygen level dependent signal (BOLD). fMRI data offers a range of insights, and can be roughly divided into two categories:

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Journals on medical image computing Medical Image Analysis (MedIA) ; also the official journal of The MICCAI Society, which organizes the Annual MICCAI Conference a premier conference for medical image computing IEEE Transactions on Medical Imaging (IEEE TMI) Medical Physics Journal of Digital Imaging (JDI) ; the official journal of the Society of Imaging Informatics Computerized Medical Imaging and Graphics Journal of Computer Aided Radiology and Surgery BMC Medical Imaging In addition the following journals occasionally publish articles describing methods and specific clinical applications of medical image computing or modality specific medical image computing

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Atlas-based segmentation: For many applications, a clinical expert can manually label several images; segmenting unseen images is a matter of extrapolating from these manually labeled training images. Methods of this style are typically referred to as atlas-based segmentation methods. Parametric atlas methods typically combine these training images into a single atlas image, while nonparametric atlas methods typically use all of the training images separately. Atlas-based methods usually require the use of image registration in order to align the atlas image or images to a new, unseen image. **Shape-based segmentation:** Many methods parametrize a template shape for a given structure, often relying on control points along the boundary. The entire shape is then deformed to match a new image. Two of the most common shape-based techniques are active shape models and active appearance models. These methods have been very influential, and have given rise to similar models. **Image-based segmentation:** Some methods initiate a template and refine its shape according to the image data while minimizing integral error measures, like the active contour model and its variations. **Interactive segmentation:** Interactive methods are useful when clinicians can provide some information, such as a seed region or rough outline of the region to segment. An algorithm can then iteratively refine such a segmentation, with or without guidance from the clinician. **Manual segmentation,** using tools such as a paint brush to explicitly define the tissue class of each pixel, remains the gold standard for many imaging applications. Recently, principles from feedback control theory have been incorporated into segmentation, which give the user much greater flexibility and allow for the automatic correction of errors. **Subjective surface segmentation:** This method is based on the idea of evolution of segmentation function which is governed by an advection-diffusion model. To segment an object, a segmentation seed is needed (that is the starting point that determines the approximate position of the object in the image). Consequently, an initial segmentation function is constructed. The idea behind the subjective surface method is that the position of the seed is the main factor determining the form of this segmentation function. **Convolutional neural networks (CNNs):** The computer-assisted fully automated segmentation performance has been improved due to the advancement of machine learning models. CNN based models such as SegNet, UNet, ResNet, AATSN, Transformers and GANs have fastened the segmentation process. In the future, such models may replace manual segmentation due to their superior performance and speed. There are other classifications of image segmentation methods that are similar to categories above. Another group, which is based on combination of methods, can be classified as "hybrid".